Experiment - 11

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(A)
A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table.
Write an SQL query to find the employees whose salary has increased in every recorded month compared to the immediately previous month. For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

The output should display:
Employee ID
Employee Name

Output:

employee_id	employee_name
101	Alok
103	Raj

Solution:

```
SELECT employee_id, employee_name
FROM (
   SELECT *,
        LAG(salary) OVER(
```

```
        PARTITION BY employee_id

        ORDER BY salary_month
    ) prev_salary
FROM EmployeeSalaryHistory


) E
GROUP BY employee_id, employee_name
HAVING COUNT(*) FILTER (
    WHERE salary <= prev_salary
) = 0;
```

(B)

A company maintains employee details in an Employee table and department information in a Department table.

Write an SQL query to find the employees whose salaries belong to the top three distinct salary levels within their respective departments. If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

The output should display:

Department Name

Employee Name

Salary

Output:

Department	Employee	Salary
IT	Max	90000
IT	Jim	85000
IT	John	85000
IT	Alex	75000
Sales	Henry	80000
Sales	Susan	75000
Sales	David	70000
HR	Tina	60000
HR	Rita	55000
HR	Manoj	50000

Solution:

```
SELECT D.NAME AS DEPARMENT, E.NAME AS EMPLOYEE, E.SALARY
FROM (


        SELECT *,
            DENSE_RANK() OVER(PARTITION BY DEPARTMENTID ORDER BY
SALARY DESC) AS RNK
        FROM EMPLOYEE
) AS E
JOIN DEPARTMENT AS D
ON E.DEPARTMENTID = D.ID
WHERE RNK<=3;
```